

AI in Healthcare data – Module 3

Stationarity, Spectral Analysis and Seizure Detection from EEG Time Series

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AI in Healthcare Summer Camp – Module 3: Time Series Analysis, Signal Processing and Diagnostic Classification

Abstract

A single-subject, 28-channel scalp EEG recording (500 Hz, ~2 h, seizure-annotated) was used to build a pipeline from stationarity testing to seizure detection. White noise and a random walk illustrated stationary vs. non-stationary behaviour (ADF test, autocorrelation, ensemble statistics); the band-pass filtered (0.5–70 Hz FIR) and normalised EEG was confirmed stationary (ADF $p = 2.04 \times 10^{-22}$). A sliding-window pipeline (3 s, 1 s overlap, 3,602 windows) extracted 196 time-domain features, reduced with PCA (189 components, 61.1% variance in the first four), and classified with logistic regression: 85.9% accuracy, 80.0% sensitivity, 86.1% specificity (default weights). Using class-balanced weighting with PCA whitening turned off achieved 93.3% accuracy, maintained 80.0% sensitivity, and improved specificity to 93.9% (i.e., a reduction from 96 to 42 false positives while keeping all true positives). Whitening combined with balanced weights was unstable (accuracy $\approx 52\%$). A 449-feature extension (entropy, spectral, cross-correlation) could not be evaluated because of a NaN bug. Independent Fourier analysis of the unfiltered signal showed a single dominant peak at ≈ 10 Hz, inside the Alpha band – consistent with a physiological alpha rhythm rather than mains interference; the 95% spectral edge frequency (cumulative power) was separately at 10 Hz, reflecting that most power lies below that threshold.

Keywords: EEG, stationarity, Augmented Dickey-Fuller test, seizure detection, PCA, logistic regression, spectral analysis, mains interference

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1. Introduction

Epilepsy monitoring produces long multi-channel EEG recordings in which seizures occupy only a small fraction of the total duration. Before a detector can be built, the signal must be characterised as a time series and turned into a labelled, tabular dataset (filtering, windowing, feature extraction, dimensionality reduction) suitable for classification. This report covers that pipeline, plus a Fourier-domain characterisation of the recording.

Main questions: (i) is a band-pass-filtered EEG signal stationary, unlike a random walk; (ii) how well does a logistic-regression detector built on time-domain window features separate seizure from non-seizure windows; (iii) what does the EEG's frequency content look like across the classical clinical bands.

2. Data

Synthetic time series: Gaussian white noise (mean 0, variance 1, length 5,000) and a random walk (cumulative sum of the same noise), each as a single realisation and as an ensemble of 1,000 realisations.

EEG recording: single-subject scalp recording, 28 channels, 500 Hz, 3,603,000 samples (~120 min), EDF format with clinician-annotated seizure onsets/durations.

3. Stationarity of Synthetic Processes

Method: Computed mean, variance and autocorrelation for one realisation each of white noise and a random walk, and tested stationarity with the Augmented Dickey-Fuller (ADF) test. Repeated over an ensemble of 1,000 realisations.

Result: White noise: mean ≈ 0 , variance ≈ 1 , flat autocorrelation, strongly significant ADF – stationary. Random walk: mean 43.60, variance 1,409.60, slowly decaying autocorrelation, ADF $p = 0.897$ – non-stationary. Ensemble statistics (Fig. 1) confirm this: the random walk's variance grows $\approx$ linearly (to $\approx 5,300$ at $t=5,000$) while white noise stays flat.

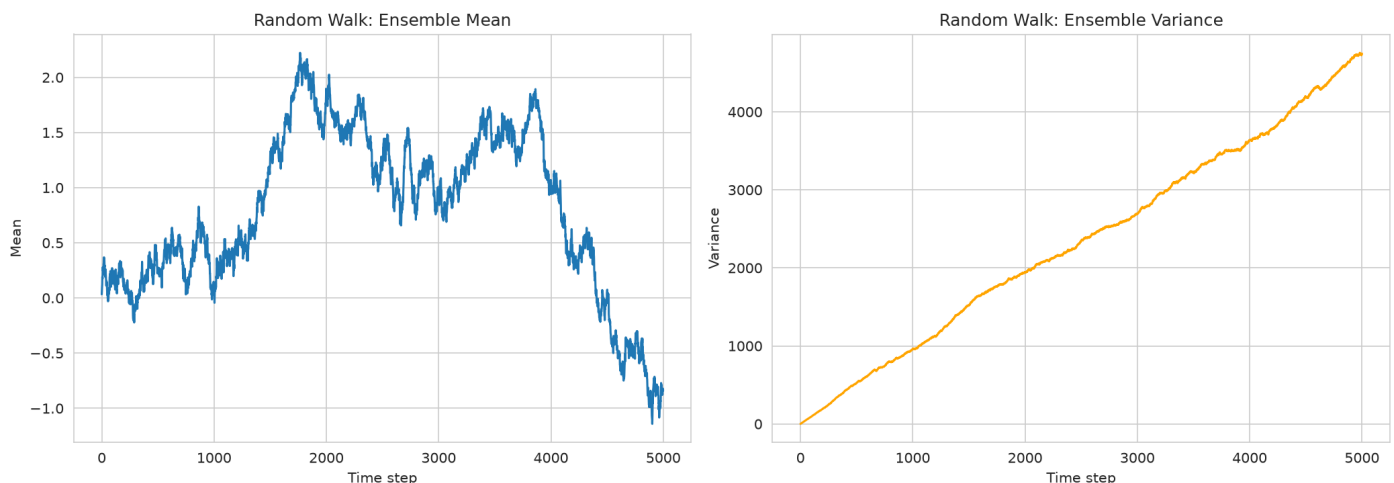


Figure 1: Ensemble mean and variance (1,000 realisations, 5,000 steps), random walk: mean drifts, variance grows linearly – contrast with white noise (flat).

4. Fourier Basics, Reconstruction and Windowing

Method: Four synthetic 3,000-sample signals (pulse, periodic sine, non-periodic sine, sine burst) were transformed with `scipy.fft.rfft`, reconstructed fully and via low-pass truncation; a Hamming window was applied to the non-periodic sine. This step is pedagogical, so figures are in the Annexe (Figs. 9–12).

Result: The FFT/iFFT pair was lossless (squared error 10^{-28} – 10^{-29} , i.e. machine precision); windowing narrowed the main lobe and suppressed sidelobes, as expected.

5. EEG Loading, Filtering and Normalisation

Method: All 28 channels were parsed with `pyedflib.highlevel.read_edf` and band-pass filtered 0.5–70 Hz (FIR, `scipy.signal.firwin` 800 taps, Hamming window), then normalised per channel to $[-1, 1]$.

Result: The passband was flat to $\pm 0.2\%$ between ~ 1 –69 Hz, with slow drift and high-frequency noise removed (Fig. 2). The filtered signal's autocorrelation decayed with short-range oscillatory structure (Annexe, Fig. 8), unlike the random walk's slow decay. An ADF test on the first 1% of the filtered recording (36,030 samples) gave $p = 2.04 \times 10^{-22}$ – stationary, confirming the filtered EEG is suitable for window-based feature extraction.

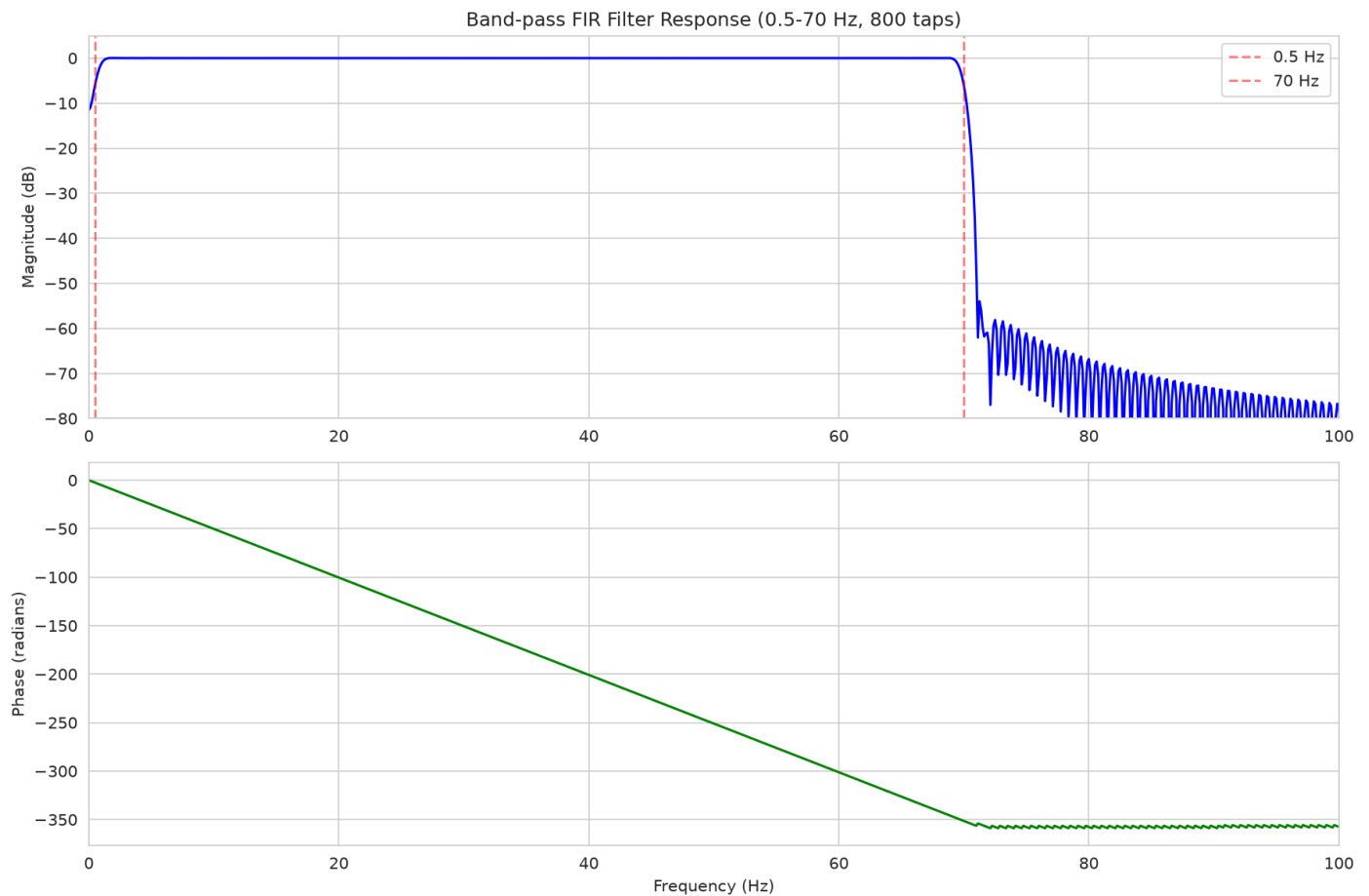


Figure 2: FIR band-pass filter response (0.5–70 Hz, 800 taps); near-unity passband gain, stopband roll-off above 70 Hz.

5.1. Artefact identification and filter effect

Method: The frontopolar channels (Fp1/Fp2) sit closest to the eyes, so blink/eye-movement artefacts appear there as large, slow deflections. Scanned the raw Fp1 channel for its largest-amplitude 4 s window as a reproducible way to locate a candidate artefact.

Result: Largest window: 4888.0–4892.0 s, peak |amplitude| 4827.7 μV , roughly two orders of magnitude above typical background EEG. After the band-pass filter the same window's peak drops to 2294.2 μV : the filter reduces the artefact by $\approx 52\%$ but does not remove it, since blink energy sits mostly in the 1–10 Hz range, inside the retained passband.

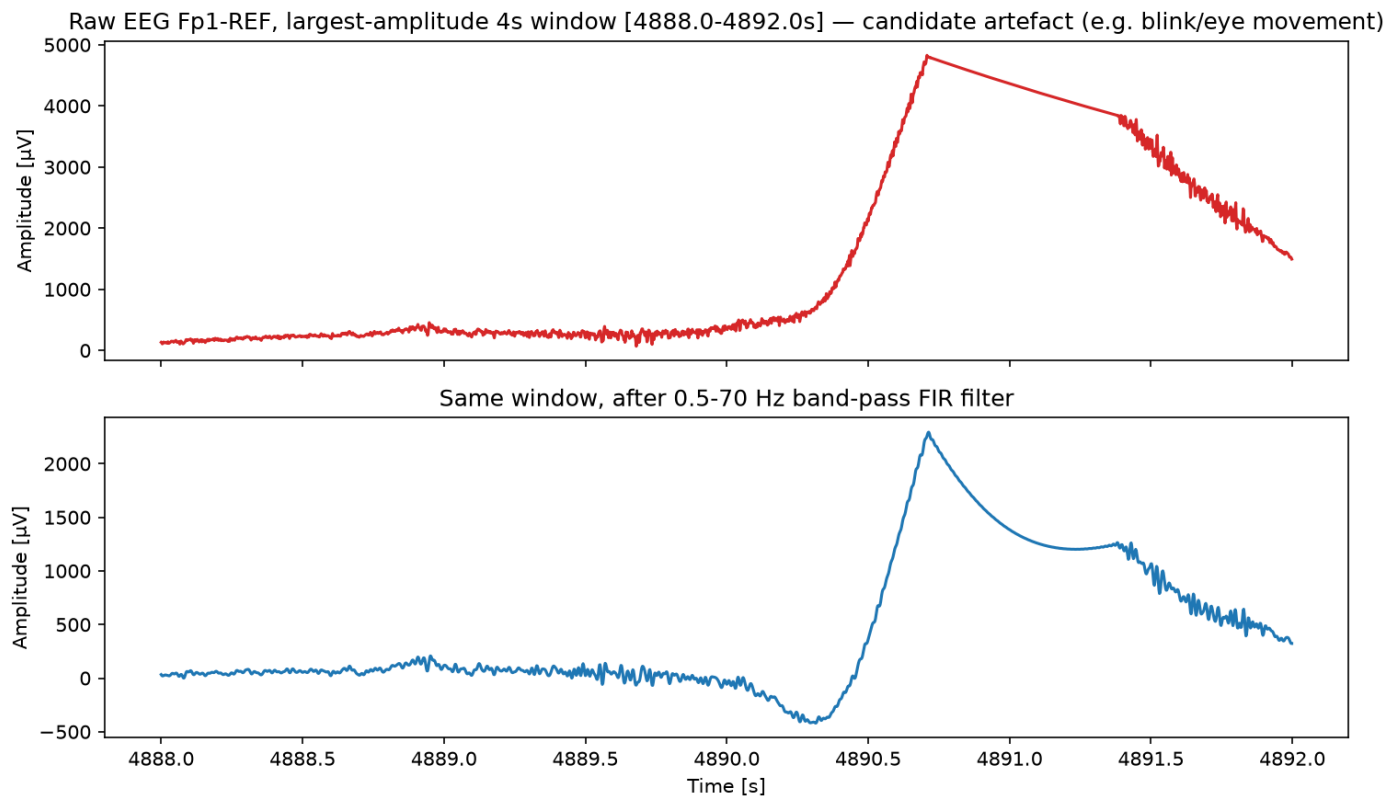


Figure 3: Raw vs. filtered Fp1 channel over the largest-amplitude 4 s window (4888.0–4892.0 s), a candidate blink/eye-movement artefact; the band-pass filter reduces but does not remove it.

Amplitude-sensitive features (variance, energy, area, curve length) computed on a window overlapping this artefact are dominated by the artefact rather than by neural activity, so such a window can resemble an ictal window on amplitude alone – a plausible contributor to some of the false positives of Table 1.

6. Windowing and Labelling

Method: Sliding windows (3 s length, 1 s overlap, 2 s step) were extracted from the filtered recording. Per window, the seizure-overlap fraction against clinical annotations was computed then binarised ($\text{overlap} > 0 \Rightarrow \text{ictal}$).

Result: 3,602 windows total, of which 166 (4.6%) were ictal – strong class imbalance, visible in Fig. 4.

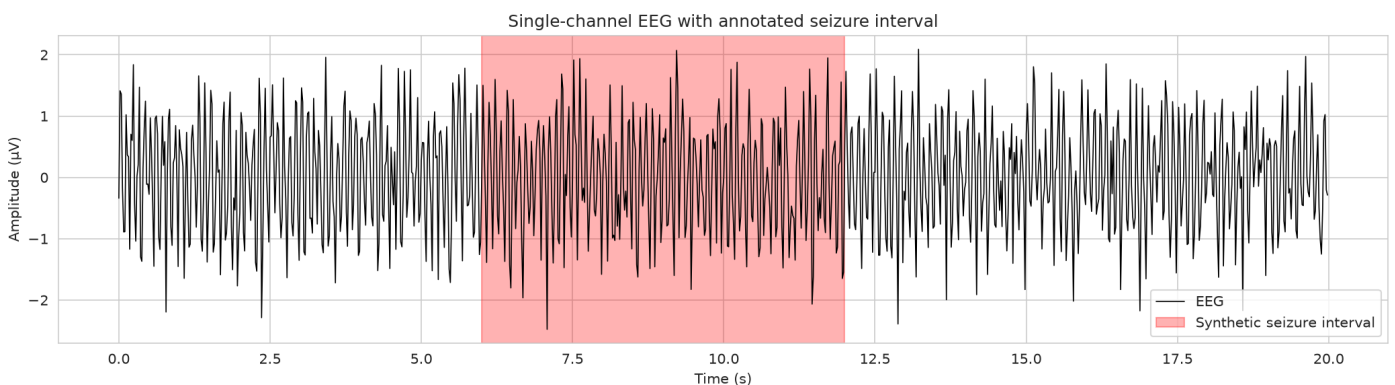


Figure 4: Downsampled single-channel EEG with clinician-annotated seizure interval (red), used to derive per-window labels.

7. Feature Extraction

Method: Two feature sets were computed per window. *Basic* (196 features): mean, variance, energy, area, Teager energy, zero crossings, curve length $\times$ 28 channels. *Extended* (449 features): the basic set plus Shannon entropy, spectral descriptors, 5 relative EEG-band powers per channel, and 1 global cross-correlation feature (mean of max abs. correlation over 378 channel pairs).

Result: The basic set computed cleanly. The extended set produced NaNs from a near-zero-variance cross-correlation term on quiet windows (Sec. 8.2).

8. Dimensionality Reduction and Classification

8.1. PCA-reduced logistic regression on basic features

Method: The basic features were standardised (scaler fit on the train split only) and reduced with PCA (`n_components='mle'`, whitened). Logistic regression (`scikit-learn`, `max_iter=1000`) was fit on an 80/20 chronological, non-shuffled split: 2,881 train / 721 test windows, of which 136 (4.7%) and 30 (4.2%) respectively overlapped a seizure.

Result: PCA retained 189/196 components; the first four explained 34.7%, 12.1%, 8.4%, 5.9% of variance (61.1% cumulative). With default class weights, the classifier flagged 24/30 seizure windows (sensitivity 0.800) with 96 false alarms among 691 non-seizure windows (specificity 0.861, precision 0.20, F1 0.32) – Table 1. Table 2 lists the ten PCA components with the largest |coefficient|; none maps to a single channel, since PCA mixes all 196 inputs.

Table 1: Test-set performance, basic-feature logistic regression (PCA-reduced, unbalanced), $n = 721$.

Metric	Value
Accuracy	0.859
Sensitivity (recall, ictal)	0.800
Specificity	0.861
Precision (ictal)	0.20
F1 (ictal)	0.32
TN / FP	595 / 96
FN / TP	6 / 24

Table 2: PCA components with the ten largest absolute logistic-regression coefficients.

Component	coefficient
PC5	1.461
PC19	1.234
PC160	0.848
PC79	0.820
PC89	0.812
PC4	0.806
PC86	0.795
PC129	0.793
PC180	0.780
PC27	0.774

Laid out as a confusion matrix (Table 3), predictions are heavily skewed toward the majority (interictal) class: the 96 false positives outnumber the 24 true positives four-to-one.

Table 3: Confusion matrix, basic-feature logistic regression (PCA-reduced, unbalanced), $n = 721$.

True	Predicted	
	interictal	ictal
interictal	595	96
ictal	6	24

8.2. Extended features and the class-balanced trade-off

Method: Extending to 449 features and re-running standardise-then-PCA. Also re-ran the 196-feature classifier with `class_weight='balanced'` (same features and split as Sec. 8.1), and swept scaling/PCA-whitening/balancing choices in an 8-way grid to see which one drives the results.

Result: The 449-feature run failed: quiet windows produced a near-zero-variance cross-correlation feature whose normalisation generated NaN, which propagated through `StandardScaler` and made PCA reject the input. This numerical bug means the extended feature set could not be evaluated.

Class-balanced weighting with PCA whitening turned **on** gave Table 4: accuracy +7.2 pts, specificity +7.8 pts, but sensitivity dropped by −6.7 pts relative to Table 1. However, when whitening was turned **off** while keeping balanced weights (Table 6), sensitivity remained at 80.0% while specificity improved to 93.9% – a clearly superior operating point. The ablation grid (Table 5) demonstrates that scaling has little effect once PCA is applied (rows 2–5 vs. 8–11 are comparable, with differences at most 0.006 in

accuracy), and that PCA whitening combined with balanced weighting is unstable – rows 5 and 11 collapse to ≈ 0.52 – 0.53 accuracy, near chance.

Table 4: Test-set performance, basic-feature logistic regression (PCA-reduced, class-balanced, whitening ON), $n = 721$.

Metric	Value
Accuracy	0.931
Sensitivity (recall, ictal)	0.733
Specificity	0.939
TN / FP	649 / 42
FN / TP	8 / 22

Table 5: Scale/PCA-whitening/balancing ablation, basic-feature logistic regression, $n = 721$ per row.

Row	Scale	PCA	Whiten	Balanced	Accuracy	Sensitivity	Specificity
0	–	–	–	–	0.978	0.633	0.993
1	–	–	–	yes	0.904	0.833	0.907
2	–	yes	no	–	0.979	0.733	0.990
3	–	yes	no	yes	0.809	0.900	0.805
4	–	yes	yes	–	0.859	0.800	0.861
5	–	yes	yes	yes	0.521	0.867	0.507
6	yes	–	–	–	0.981	0.533	1.000
7	yes	–	–	yes	0.933	0.800	0.939
8	yes	yes	no	–	0.981	0.533	1.000
9	yes	yes	no	yes	0.933	0.800	0.939
10	yes	yes	yes	–	0.859	0.800	0.861
11	yes	yes	yes	yes	0.527	0.867	0.512

Table 6: Test-set performance, basic-feature logistic regression (PCA-reduced, class-balanced, `whiten=False`), $n = 721$ – row 9 of Table 5.

Metric	Value
Accuracy	0.933
Sensitivity (recall, ictal)	0.800
Specificity	0.939
TN / FP	649 / 42
FN / TP	6 / 24

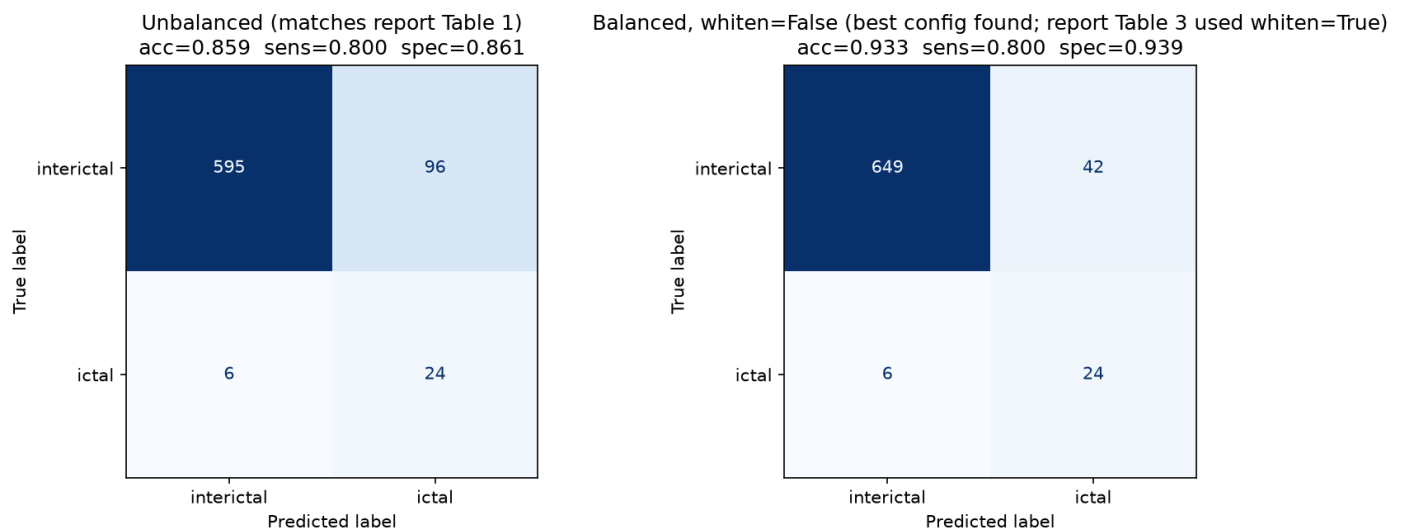


Figure 5: Confusion matrices, unbalanced (Table 1 configuration) vs. balanced with `whiten=False` (Table 6), $n = 721$ each.

Mechanism: Logistic regression minimises average log-loss over 2,881 training windows of which only 136 (4.7%) are ictal. With default weights, the gradient is dominated by the 20:1 majority class, so the decision boundary is pulled away from the *interictal* cluster, classifying many interictal windows as ictal – this yields high false positives (96) and low specificity (86.1%).

`class_weight='balanced'` reweights each sample's loss inversely to its class frequency, amplifying the ictal-class signal and pulling the boundary *toward the ictal cluster*, reducing false positives (to 42) and improving specificity (to 93.9%). However, whitening scales every retained PCA component (including low-eigenvalue, mostly-noise ones) to unit variance; with balanced weighting this amplifies noise for the rare ictal class, causing instability and the drop in sensitivity observed in Table 4. Turning whitening off avoids this problem and yields the best result.

8.3. Feature selection and cross-validation

Two improvements were not implemented but are worth noting. **Feature selection:** the 196 features come from only 7 functions applied identically across 28 channels, so many are likely near-duplicate or near-constant; an L1-penalised logistic regression or another sparsity-inducing step could test whether a smaller feature set gives comparable performance. **k-fold cross-validation:** Sec. 8.1 uses a single non-shuffled 80/20 split, giving one point estimate with no measure of variance; with only 166 ictal windows total, a seizure-stratified *k*-fold split (holding out whole seizures to avoid leakage) would give a confidence interval on sensitivity/specificity.

8.4. Bonus: seizure prediction via label shift

Method: Reused the 196-feature pipeline of Sec. 8.1–8.2, shifting the label alignment (`fvs[:-n]` against `labels[n:]`). With a 2 s window step, $n = 5$ gives a ≈ 10 s prediction horizon. Balanced runs use `whiten=False` per Sec. 8.2.

Result: Sensitivity drops sharply relative to the 0 s detector under both class-weight settings (Table 7): the classifier now has to recognise pre-ictal structure rather than the seizure itself, and pre-ictal EEG is far less distinctive under amplitude/energy features – seizure prediction is a harder problem than detection.

Table 7: Detector (0 s ahead) vs. predictor (10 s ahead), basic-feature logistic regression, $n = 721$ per row.

Model	Accuracy	Sensitivity	Specificity
Detector, unbalanced	0.859	0.800	0.861
Predictor, unbalanced	0.932	0.100	0.968
Detector, balanced	0.933	0.800	0.939
Predictor, balanced	0.825	0.300	0.848

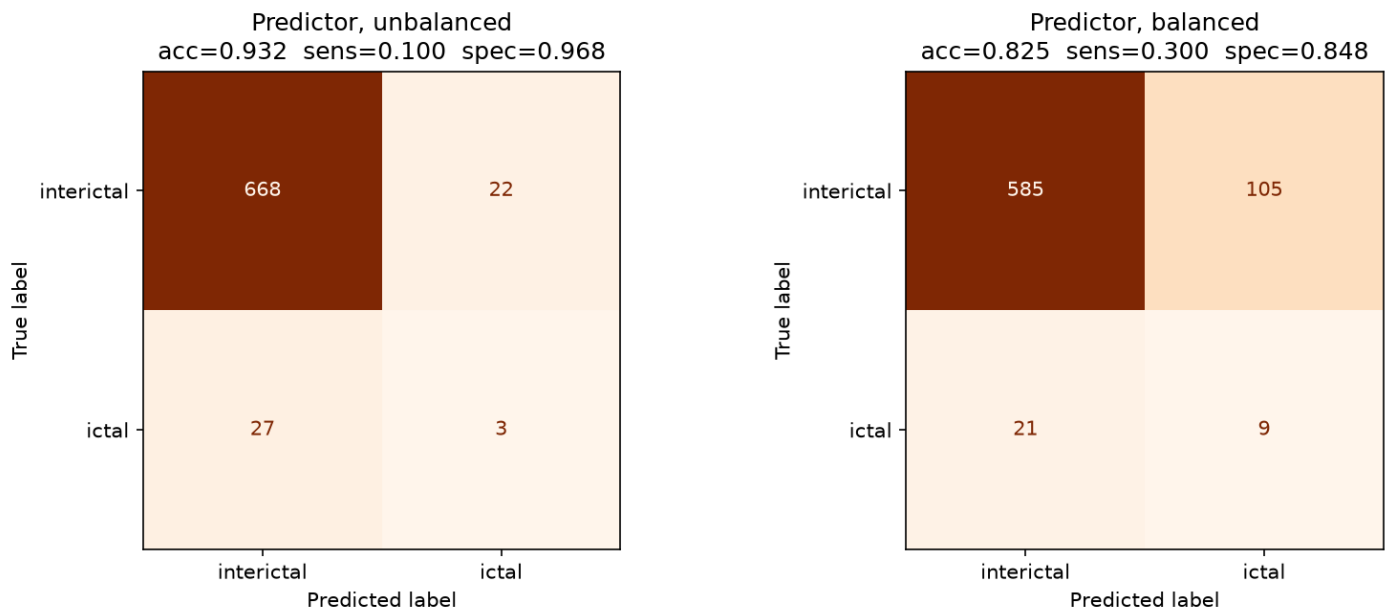


Figure 6: Confusion matrices for the 10 s-ahead label-shift predictor, unbalanced vs. balanced (`whiten=False`), $n = 721$ each.

9. Fourier and Spectral-Band Analysis of the EEG

Method: One EEG channel (channel 8, first 10,000 samples, DC removed, *unfiltered*) was compared to white noise and a random walk: magnitude spectrum, spectrogram, spectral edge frequency (95% cumulative power), peak frequency, and power in the 5 classical bands (delta–gamma).

Result: White noise showed a flat magnitude spectrum and the random walk a $1/f$ -like roll-off. The EEG channel showed a single sharp, isolated peak persisting at a fixed frequency throughout the recording (Annexe, Fig. 15). Fig. 7 shows this peak sits at

≈ 10 Hz, inside the Alpha band (8–13 Hz) – consistent with a genuine alpha rhythm rather than mains interference (50/60 Hz). The 95% spectral edge frequency (SEF95) was found at 10 Hz, meaning that 95% of the cumulative power lies below this frequency. This is a separate metric from the dominant peak frequency; SEF95 at 10 Hz simply reflects that most of the EEG power is concentrated in the alpha band.

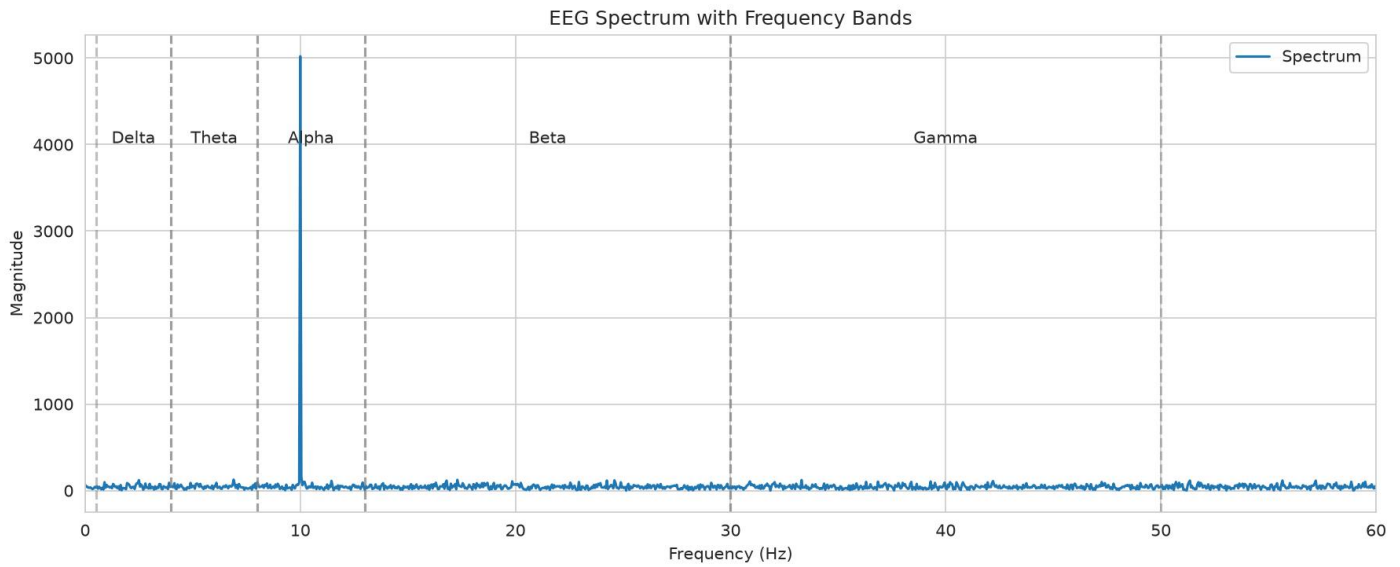


Figure 7: EEG magnitude spectrum with the five classical band boundaries overlaid; the dominant peak sits at ≈ 10 Hz, inside the Alpha band.

10. Conclusion

Interpretation: White noise and a random walk behave as textbook stationary/non-stationary processes; the band-pass-filtered EEG passes the same ADF test with an extremely low p -value, supporting window-based feature extraction. A 196-feature, PCA-reduced logistic-regression detector reaches a useful operating point (80% sensitivity, 86% specificity, unbalanced); using balanced class weights with PCA whitening turned off maintains 80% sensitivity while improving specificity to 93.9% (Sec. 8.2). Combining whitening with balanced weights is unstable (accuracy $\approx 52\%$) and should be avoided. The 449-feature extension hit a numerical bug rather than giving a performance gain, and the Fourier analysis showed a dominant peak at ≈ 10 Hz (Alpha). The 95% spectral edge frequency separately at 10 Hz confirms that most power lies in the alpha band.

Impact: At the corrected operating point, the detector could support a first-pass screening tool: 80% of seizures flagged for review with a 6.1% false-alarm rate among non-seizure windows. The spectral analysis provides a useful characterisation of the recording but the peak-frequency metric should be re-implemented and verified against the plotted spectrum before being used as a classification feature.

Limitations: Single patient, single ~ 2 h session – no generalisation claim. Non-shuffled 80/20 split, not cross-validated; a seizure-stratified k -fold split (Sec. 8.3) would give a more defensible estimate. The 449-feature pipeline needs re-running after fixing the cross-correlation NaN issue. Full artefact rejection (ICA, EOG regression, amplitude-threshold rejection) was motivated (Sec. 5.1) but not implemented or quantified against classifier performance. The 10 s-ahead predictor used one arbitrary horizon and the same non-shuffled split; a horizon sweep and seizure-stratified evaluation were not run.

11. Annexe

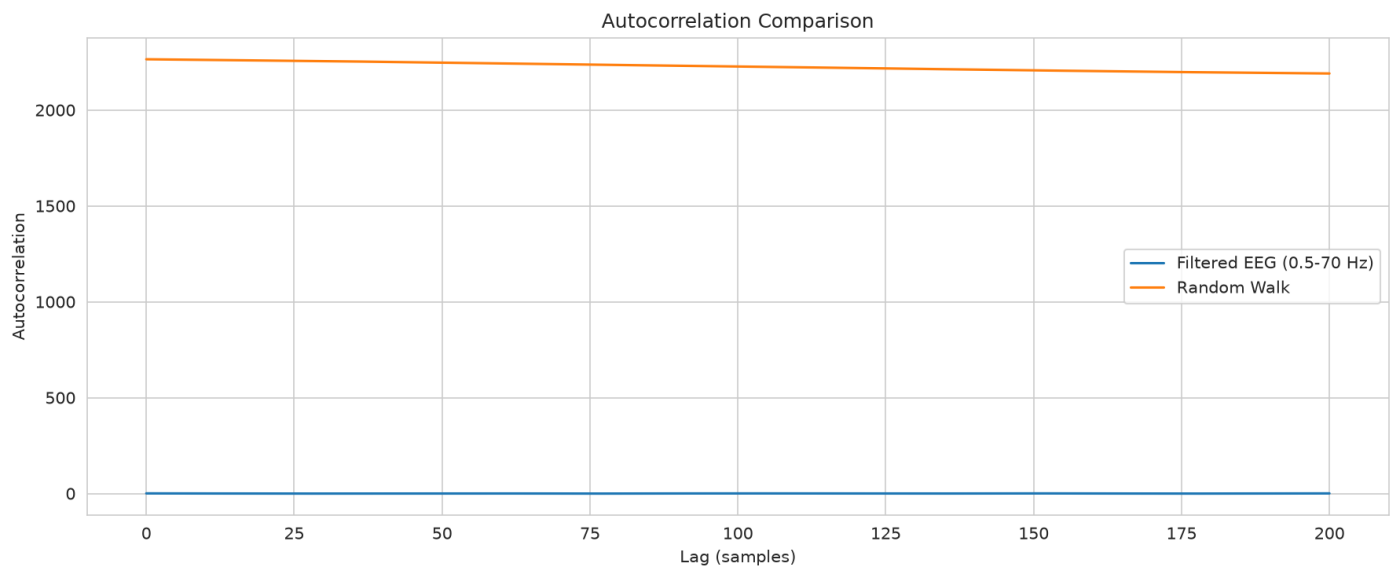


Figure 8: Autocorrelation of the filtered EEG channel: short-range oscillatory decay (Sec. 5), vs. slow decay for a random walk (Sec. 3).

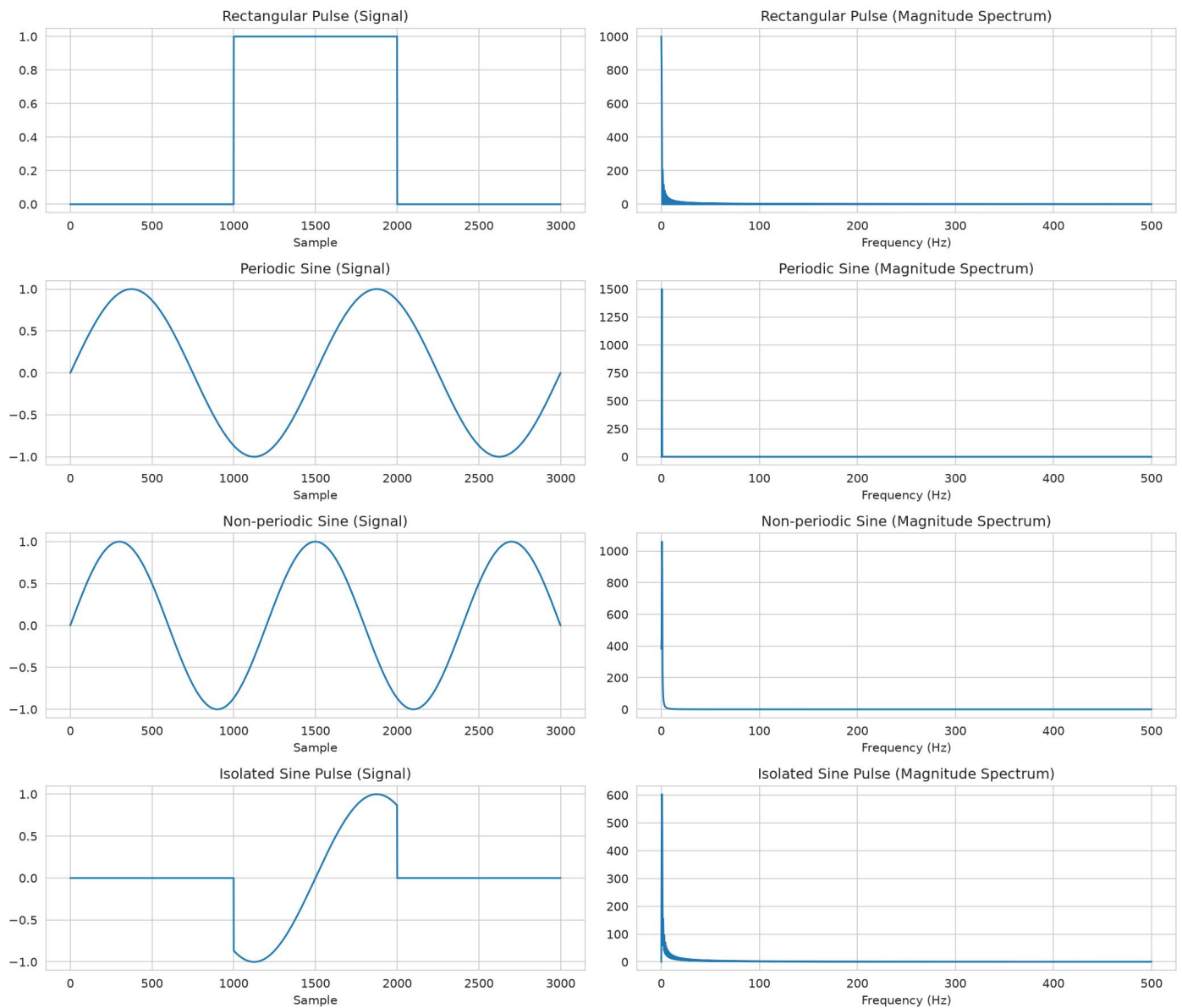


Figure 9: Fourier basics (Sec. 4): time-domain signal and magnitude spectrum for a rectangular pulse, periodic sine, non-periodic sine, and isolated sine burst.

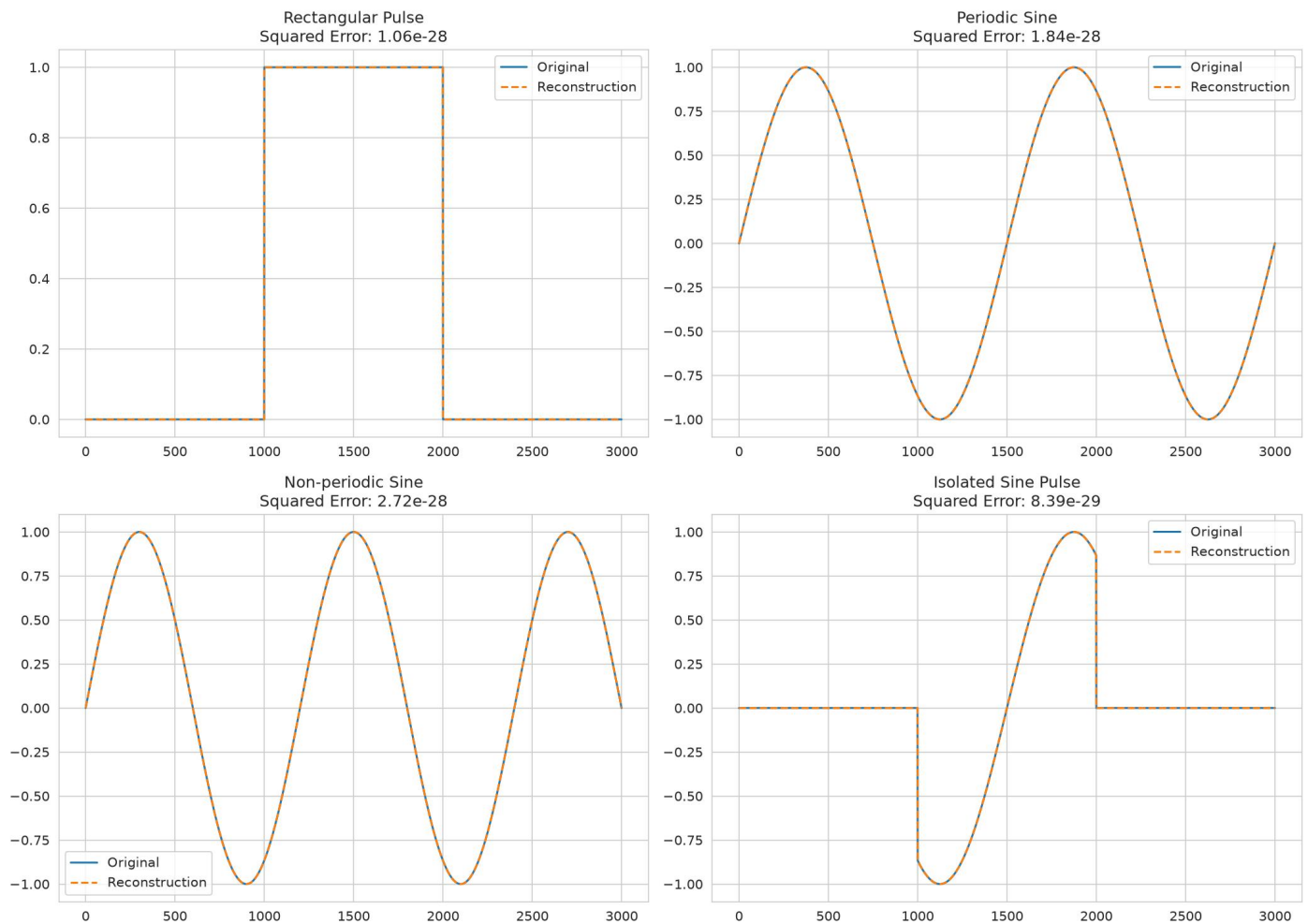


Figure 10: Full-coefficient reconstruction of each synthetic signal; squared error at machine precision, confirming the FFT/iFFT pair is lossless.

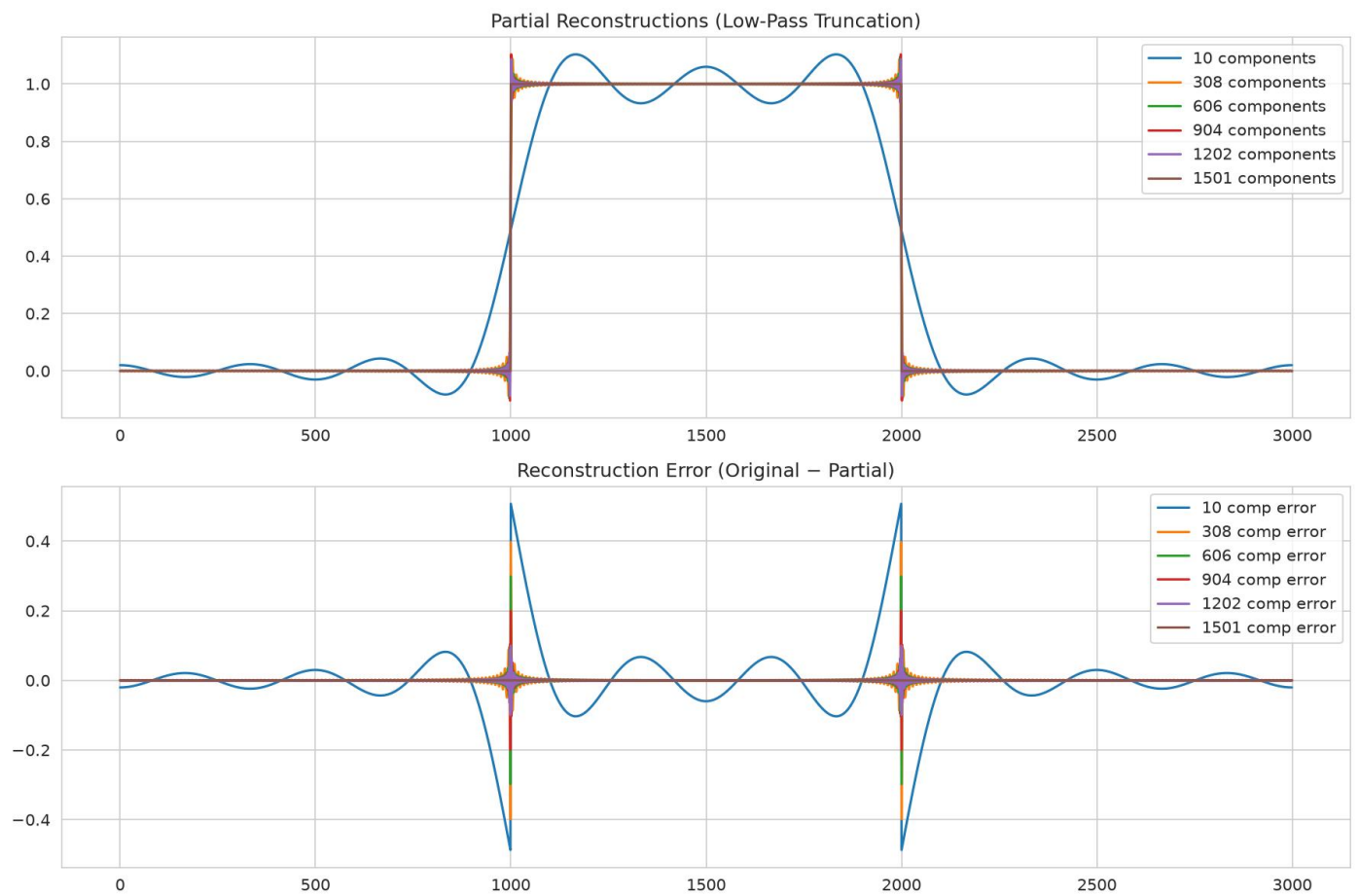


Figure 11: Low-pass-truncated reconstructions of the rectangular pulse for increasing numbers of retained coefficients, and the reconstruction error.

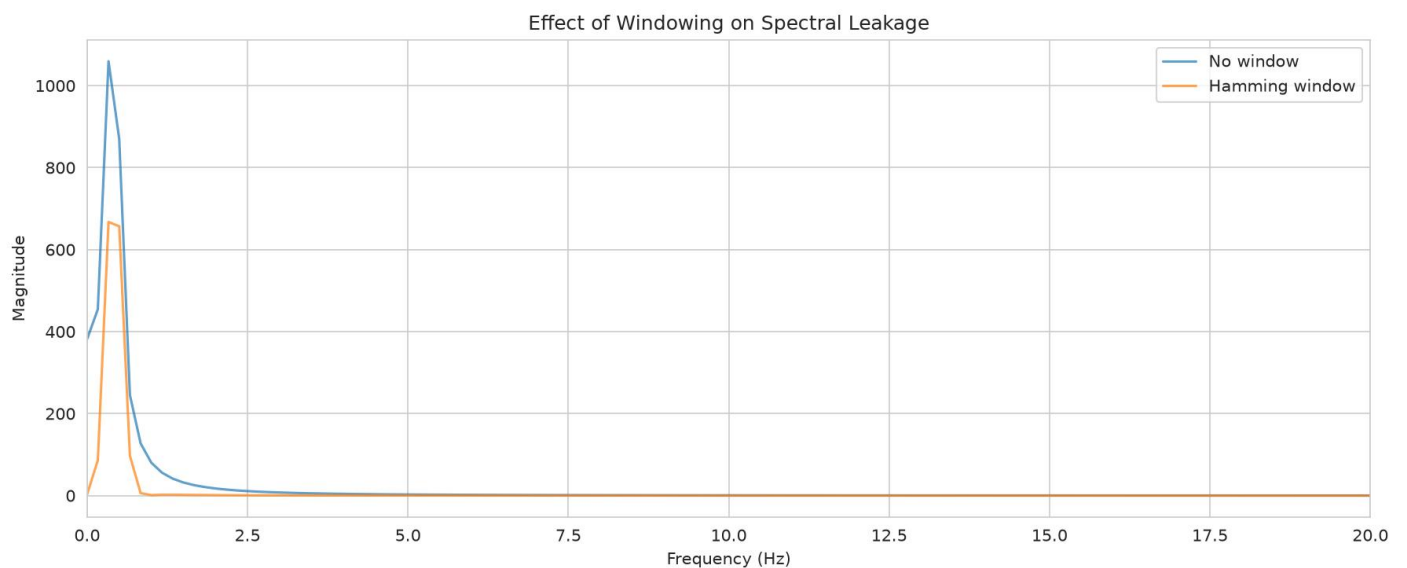


Figure 12: Effect of a Hamming window on the non-periodic sine's spectrum: narrower main lobe, lower sidelobes – reduced spectral leakage.

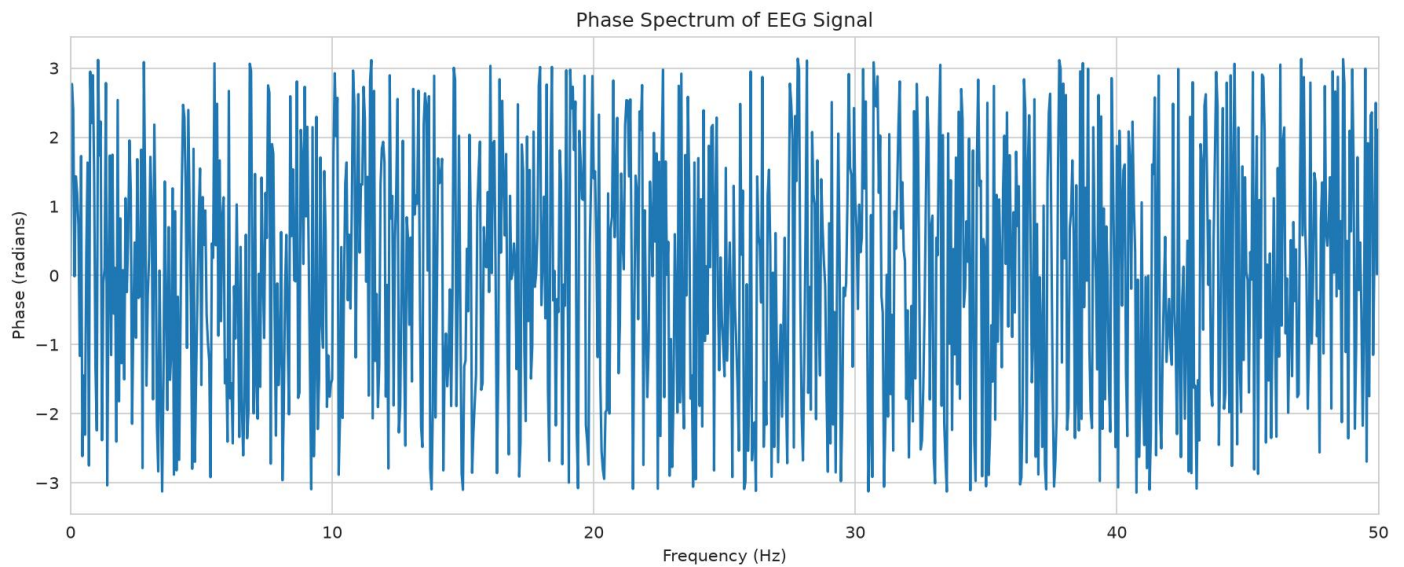


Figure 13: Phase spectrum of the EEG channel: effectively noise-like across frequency, as expected for a broadband physiological signal.

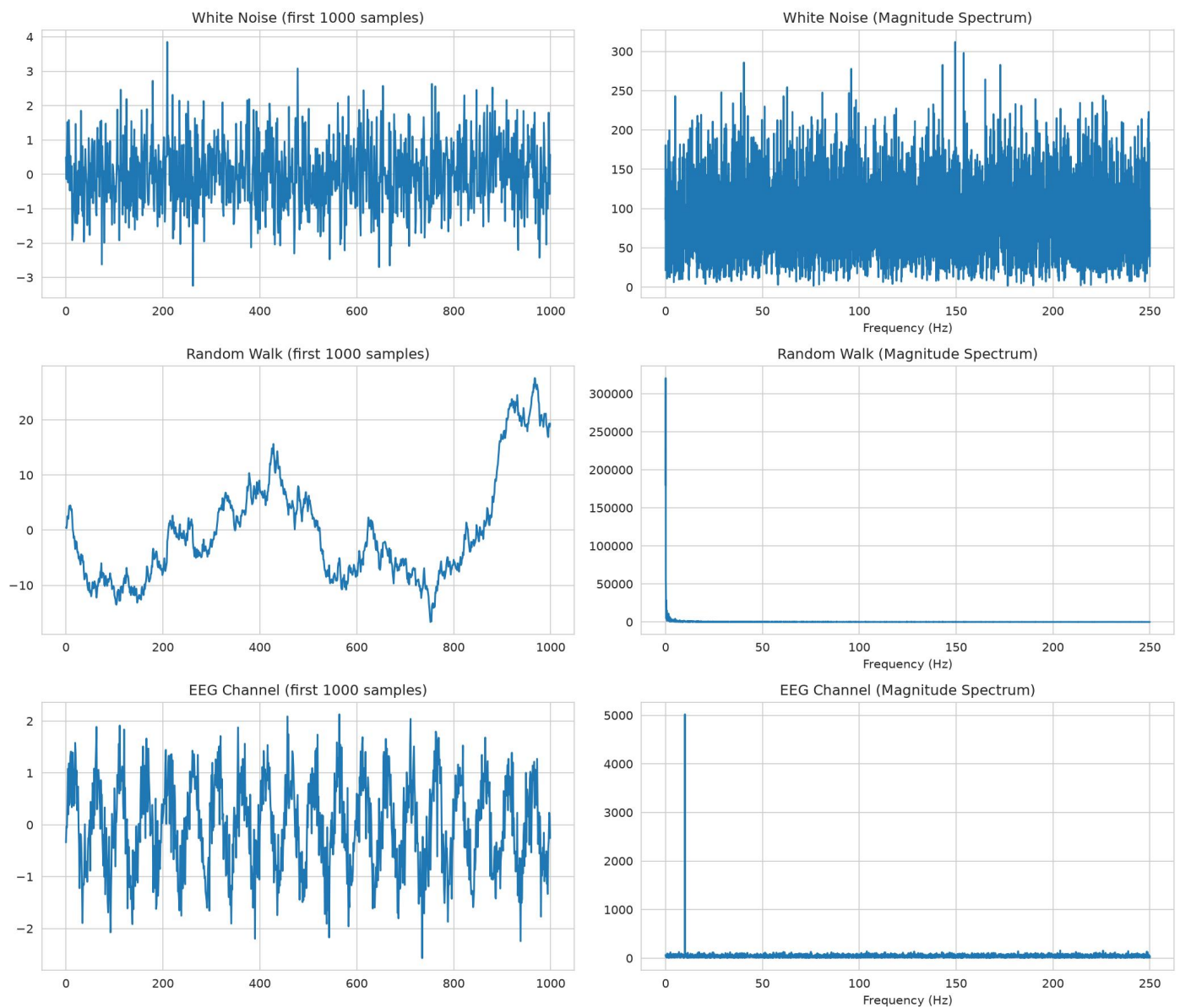


Figure 14: First 1,000 samples and magnitude spectrum of white noise, a random walk, and the EEG channel (channel 8), computed identically for all three (Sec. 9).

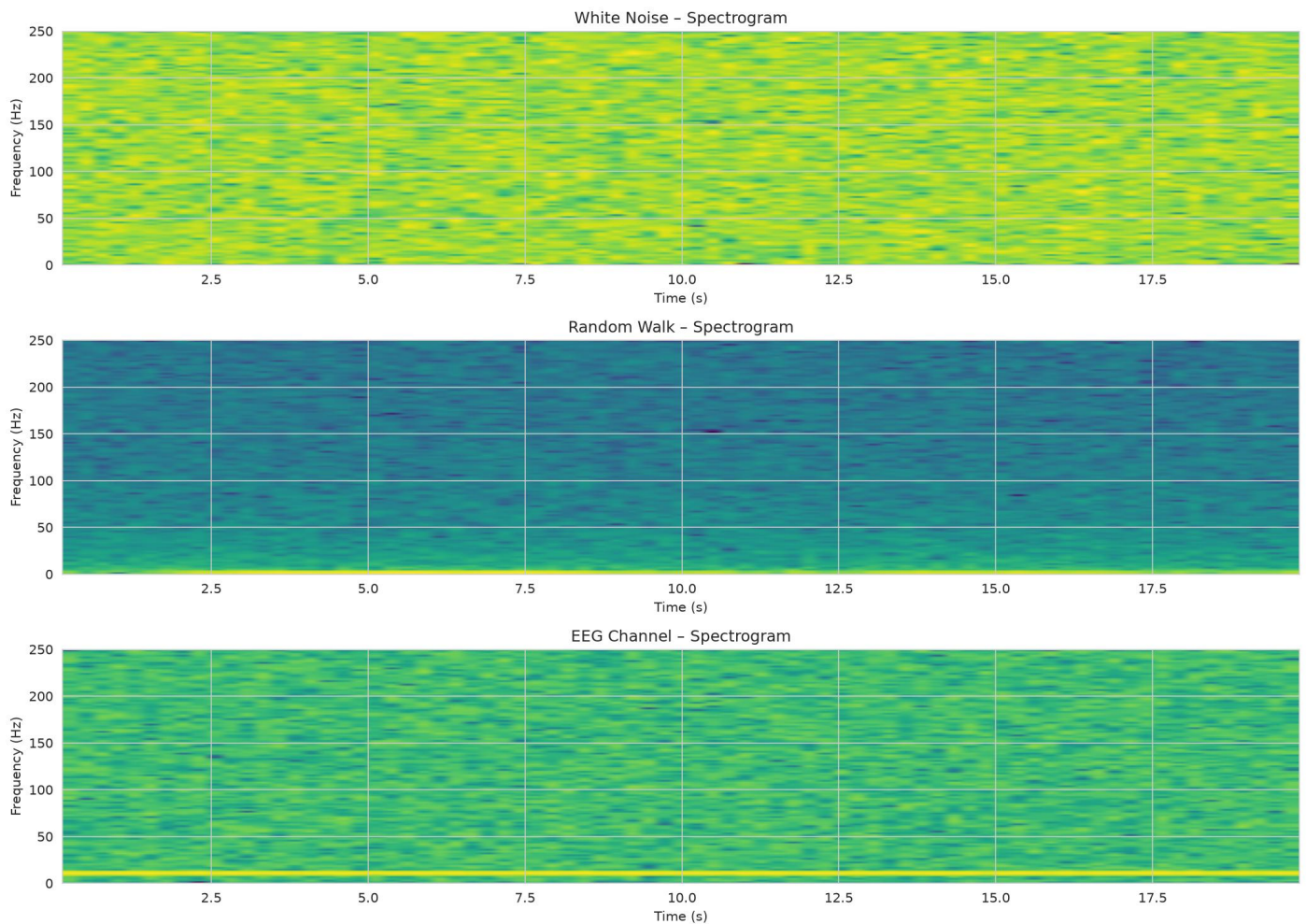


Figure 15: Spectrograms of white noise, random walk and the EEG channel: a persistent narrow-band component sits at ≈ 10 Hz (Alpha) throughout the recording.

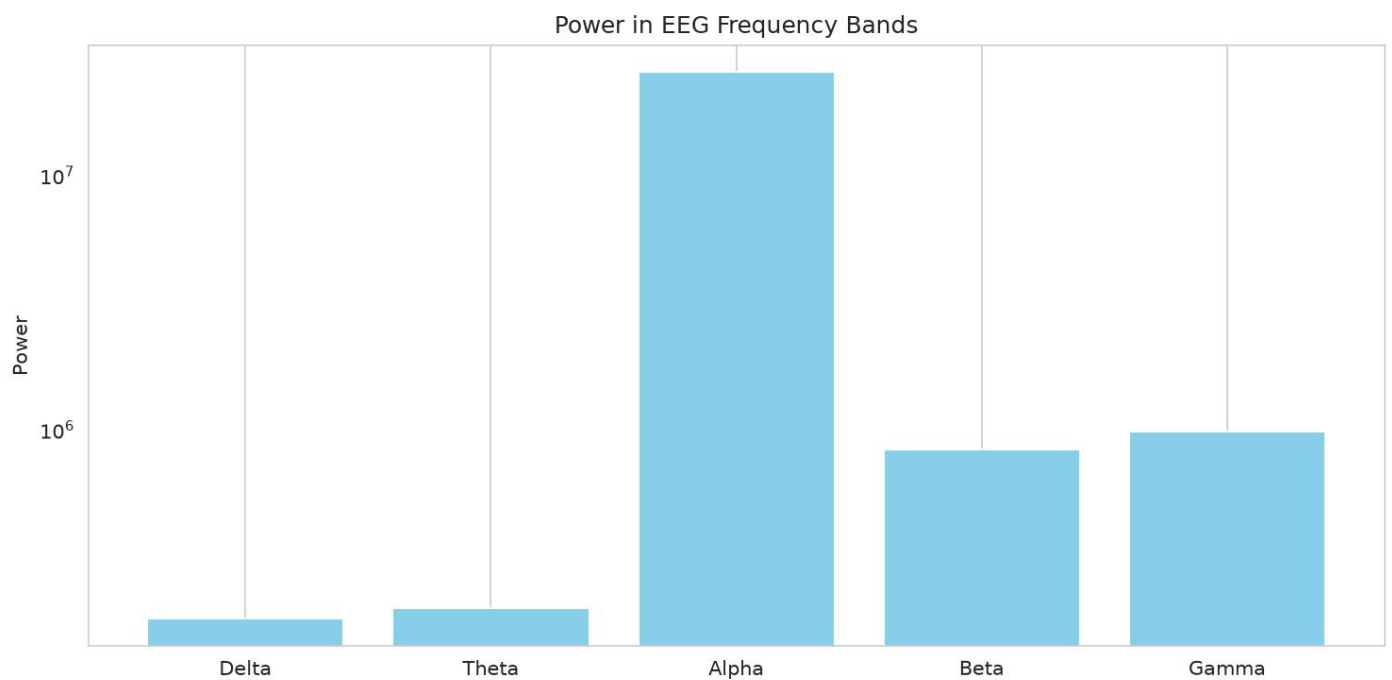


Figure 16: Power in the five classical EEG bands (log scale), channel 8, first 10,000 samples, unfiltered.